Signals & Systems (151-0575-01) Exam, February 11th, 2021 - Answer key (compact)

Condensed from the official corrected solution. 60 single-choice questions, 1 point each, all-or-nothing; calculations are not graded. Each criterion heading gives the label, name and maximum, followed by the question, the options in order (A, B, C, ...) and the key. Shared context (Box) is stated once.

Context: Consider the continuous-time LTI system G_c with transfer function H(s) = prod_{i=1}^{M} (s - b_i) / prod_{i=1}^{N} (s - a_i), where a_i, b_i in C. Consider the following discretization method, where the poles and zeros of the continuous-time system are mapped to the poles and zeros of the discretized system as follows: p_i = e^(a_i Ts) and z_i = e^(b_i Ts), where Ts is the sampling time, leading to the transfer function W(z) = prod_{i=1}^{M} (z - z_i) / prod_{i=1}^{N} (z - p_i).

1.a - MC question 1 - 1 points:

Question: Select the correct statement.

Options: A) The resulting discretized system is always stable | B) If the continuous-time system is unstable, then the discretized system is stable | C) None of the other statements is correct | D) If the continuous-time system is stable, then the discretized system is unstable | E) If the continuous-time system is stable, then the discretized system is stable | F) The resulting discretized system is always unstable

Key: option E - If the continuous-time system is stable, then the discretized system is stable

2.a - MC question 2 - 1 points:

Question: Select the correct statement.

Options: A) If the CT system is non-causal, then the discretized system is causal | B) The resulting discretized system is always causal | C) If the CT system is causal, then the discretized system is causal | D) The resulting discretized system is always non-causal | E) None of the other statements is correct | F) If the CT system is causal, then the discretized system is non-causal

Key: option C - If the CT system is causal, then the discretized system is causal

3.a - MC question 3 - 1 points:

Question: Mark the true statement.

Options: A) The transformation wraps the s-plane's j*omega axis around the z-plane's unit circle once | B) The transformation wraps the s-plane's j*omega axis around the z-plane's unit circle repeatedly | C) None of the other statements is correct | D) The transformation does not introduce frequency warping

Key: option B - The transformation wraps the s-plane's j*omega axis around the z-plane's unit circle repeatedly

4.a - MC question 4 - 1 points:

Question: Consider the continuous-time transfer function H(s) = 1 / (s + 1), mapped to the transfer function W(z) = k_d / (z + e^(-Ts)) using the aforementioned discretization technique. For which value of k_d does W(z) match the DC gain of H(s)? Hint: the DC gain is the gain at zero CT frequency, i.e. omega = 0.

Options: A) k_d = 1 + e^(-Ts) | B) k_d = -1 | C) k_d = 1 | D) k_d = 1 - e^(Ts) | E) k_d = e^(-Ts) | F) k_d = e^(Ts)

Key: option A - k_d = 1 + e^(-Ts)

Context: Consider the continuous-time signal x(t) = cos(omega t), with omega in R. [Figure: plot of x(t) for t = 0..11, amplitude between -1 and 1 - not included.] The signal is sampled with: (1) sampling period Ts1, resulting in the discrete-time signal x_1[n] = cos(pi n); (2) sampling period Ts2, resulting in the discrete-time signal x_2[n] = cos(Omega_2 n). [Figure: plot of x_2[n] for n = 0..11 - not included.]

5.a - MC question 5 - 1 points:

Question: What is the sampling time Ts1?

Options: A) Ts1 = 1 | B) Ts1 = 1/2 | C) Ts1 = 3/2 | D) Ts1 = 3/4

Key: option C - Ts1 = 3/2

6.a - MC question 6 - 1 points:

Question: What is the sampling time Ts2?

Options: A) Ts2 = 1 | B) Ts2 = 3/2 | C) Ts2 = 3/4 | D) Ts2 = 1/2

Key: option C - Ts2 = 3/4

Context: Consider the frequency response (magnitude |H(Omega)| and phase angle H(Omega), for Omega from 0 to pi) of an LTI low-pass filter designed in continuous time and discretized with sampling time Ts = 1 using three different methods (Discretization 1, 2 and 3), plotted together with the continuous-time response. [Figure not included.]

7.a - MC question 7 - 1 points:

Question: What discretization does Discretization 1 correspond to?

Options: A) Euler discretization | B) Bilinear transform | C) Exact discretization

Key: option B - Bilinear transform

8.a - MC question 8 - 1 points:

Question: What discretization does Discretization 3 correspond to?

Options: A) Bilinear transform | B) Euler discretization | C) Exact discretization

Key: option B - Euler discretization

Context: Consider the following discrete-time systems with input u and output y:
G_A: x[n+1] = 2 x[n] + u[n], y[n] = x[n]
G_B: x[n+1] = 0.5 x[n] + 2 u[n], y[n] = 2 x[n]
G_C: x[n+1] = n x[n] + u[n], y[n] = 0.5 x[n]

9.a - MC question 9 - 1 points:

Question: Select the answer containing all linear systems.

Options: A) G_A | B) G_B | C) G_C | D) G_A, G_B | E) G_A, G_C | F) G_B, G_C | G) G_A, G_B, G_C | H) No systems

Key: option G - G_A, G_B, G_C

10.a - MC question 10 - 1 points:

Question: Select the answer containing all time-invariant systems.

Options: A) G_A | B) G_B | C) G_C | D) G_A, G_B | E) G_A, G_C | F) G_B, G_C | G) G_A, G_B, G_C | H) No systems

Key: option D - G_A, G_B

11.a - MC question 11 - 1 points:

Question: Select the answer containing all causal systems.

Options: A) G_A | B) G_B | C) G_C | D) G_A, G_B | E) G_A, G_C | F) G_B, G_C | G) G_A, G_B, G_C | H) No systems

Key: option G - G_A, G_B, G_C

12.a - MC question 12 - 1 points:

Question: Select the answer containing all BIBO stable systems.

Options: A) G_A | B) G_B | C) G_C | D) G_A, G_B | E) G_A, G_C | F) G_B, G_C | G) G_A, G_B, G_C | H) No systems

Key: option B - G_B

Context: Consider the following discrete-time systems:
G_A: y[n+1] = 0.5 y[n] + u[n]
G_B: impulse response {h[n]} = {..., 0, 1 (n=0), 1/4, 3/4, 1/2, 0, 0, ...}
G_C: y[n] = u[n]^2
G_D: y[n] = sqrt(|u[n]|)
The cascade of two systems G_1 and G_2 means u -> G_1 -> y_1 = u_2 -> G_2 -> y.

13.a - MC question 13 - 1 points:

Question: Select the correct statement.

Options: A) The cascade of G_B and G_C and the cascade of G_C and G_B produce the same output. | B) The cascade of G_A and G_D and the cascade of G_D and G_A produce the same output. | C) The cascade of G_A and G_C and the cascade of G_C and G_A produce the same output. | D) The cascade of G_C and G_D and the cascade of G_D and G_C produce the same output. | E) The cascade of G_B and G_D and the cascade of G_D and G_B produce the same output. | F) None of the other statements is correct

Key: option D - The cascade of G_C and G_D and the cascade of G_D and G_C produce the same output.

Context: Let G be an LTI system with input u, output y, and step response r[n]. [Figure: stem plot of r[n] for n = -1..4, values between 0 and 2 - not included.]

14.a - MC question 14 - 1 points:

Question: What is the impulse response? (The value marked (n=0) is at time n = 0.)

Options: A) {h[n]} = {..., 0, 1 (n=0), 2, 2, 2, 0, ...} | B) {h[n]} = {..., 0, 1 (n=0), 1, 0, ...} | C) {h[n]} = {..., 0, -1, 1 (n=0), 1/2, 3/2, 0, ...} | D) None of the other statements is correct | E) {h[n]} = {..., 0, 2 (n=0), 2, 0, ...} | F) {h[n]} = {..., 0, 2, 1 (n=0), 1/3, 1/4, 0, 0, ...}

Key: option B - {h[n]} = {..., 0, 1 (n=0), 1, 0, ...}

15.a - MC question 15 - 1 points:

Question: Select the correct statement.

Options: A) The system is causal and stable. | B) The system is non-causal and unstable. | C) None of the other statements is correct | D) The system is non-causal and stable. | E) It is not possible to determine if the system is stable. | F) The system is causal and unstable.

Key: option A - The system is causal and stable.

16.a - MC question 16 - 1 points:

Question: Let G be an LTI system with input u, output y, and impulse response {h[n]} = {..., 0, 0 (n=0), 6, 6/10, 6/100, ...}. Which of the following is a state representation of G?

Options: A) A = 0.2, B = 2, C = 3, D = 0 | B) A = 0.5, B = 1, C = 4, D = 1 | C) None of the other statements is correct | D) A = 0.1, B = 2, C = 3, D = 0 | E) A = 0.1, B = 4, C = 3, D = 1 | F) A = 0.5, B = 1, C = 5, D = 0

Key: option D - A = 0.1, B = 2, C = 3, D = 0

Context: Consider the discrete-time LTI system with impulse response h[n] = 0 for n < 0, h[n] = 1 for n = 0, h[n] = a^n + b^n for n > 0, with a, b in R and a <= b.

17.a - MC question 17 - 1 points:

Question: For which z in C does the z-transform H(z) = sum_{n=-inf}^{inf} h[n] z^(-n) exist?

Options: A) |z| >= |b| | B) |z| > |a| | C) |z| < |b| | D) |z| >= |a| | E) |z| > |b| | F) |z| < |a| | G) |z| <= |a| | H) |z| <= |b|

Key: option E - |z| > |b|

18.a - MC question 18 - 1 points:

Question: Mark the associated difference equation.

Options: A) None of the other difference equations is the correct one | B) y[n] = ab y[n-1] - (a+b) y[n-2] + (a+b) u[n] + ab u[n-1] | C) y[n] = ab y[n-1] - (a+b) y[n-2] + 2 u[n] + (a+b) u[n-1] | D) y[n] = (a+b) y[n-1] - ab y[n-2] + (a+b) u[n] + 2 u[n-1] | E) y[n] = (a+b) y[n-1] - ab y[n-2] + 2 u[n] - (a+b) u[n-1]

Key: option E - y[n] = (a+b) y[n-1] - ab y[n-2] + 2 u[n] - (a+b) u[n-1]

19.a - MC question 19 - 1 points:

Question: Let a = 1 and b = -5. What is the output {y[n]} for the input u[n] = (-1)^n for all times n? (The value marked (n=0) is at time n = 0.)

Options: A) None of the other sequences is the correct one | B) {y[n]} = {..., -0.1 (n=0), 0.1, -0.1, 0.1, -0.1, 0.1, ...} | C) {y[n]} = {..., 0.1 (n=0), 0.1, 0.1, 0.1, 0.1, 0.1, ...} | D) {y[n]} = {..., 0.25 (n=0), -0.25, 0.25, -0.25, 0.25, -0.25, ...} | E) {y[n]} = {..., -0.24 (n=0), 0.12, -0.06, 0.03, -0.015, 0.007, ...} | F) {y[n]} = {..., 0.25 (n=0), 0.25, 0.25, 0.25, 0.25, 0.25, ...}

Key: option D - {y[n]} = {..., 0.25 (n=0), -0.25, 0.25, -0.25, 0.25, -0.25, ...}

Context: Given a transfer function H(z) with poles at z_p = 0.5, z_p = 0.75 and z_p = 1.5 and a zero at z_z = 0 (a pole z_p satisfies H(z_p) = infinity, a zero z_z satisfies H(z_z) = 0).

20.a - MC question 20 - 1 points:

Question: What is the largest region of z for which the transfer function H(z) is defined?

Options: A) |z| < 1.5 | B) |z| > 1.5 | C) 0.75 < |z| < 1 | D) 0.5 < |z| < 1.5 | E) 0.75 < |z| | F) 0.75 < |z| < 1.5

Key: option F - 0.75 < |z| < 1.5

Context: Consider the signal x[n]. [Figure: stem plot of x[n] for n = -3..6, values between 0 and 3 - not included.]

21.a - MC question 21 - 1 points:

Question: Which of the following plots corresponds to x[-2n + 1]? [Answer options are four plots - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option A - Plot A

22.a - MC question 22 - 1 points:

Question: Which of the following statements regarding time operations on the signal x[n] is correct?

Options: A) The signal x[n] must be an output of a time-invariant system. | B) A composition of time operations should follow a specific order. | C) None of the other statements is correct | D) The signal x[n] must be an output of a linear and time-invariant system.

Key: option C - None of the other statements is correct

Context: Consider the following causal, discrete-time LTI systems G_1 and G_2 with state-space representations:
G_1: q_1[n+1] = A_1 q_1[n] + B_1 u_1[n], y_1[n] = C_1 q_1[n] + D_1 u_1[n]
G_2: q_2[n+1] = A_2 q_2[n] + B_2 u_2[n], y_2[n] = C_2 q_2[n] + D_2 u_2[n]
G_3 is the series (cascade) connection u_3 -> G_1 -> G_2 -> y_3, with u_3 = u_1, y_1 = u_2, y_3 = y_2, q_3 = [q_1; q_2]. G_4 is the parallel connection with u_4 = u_1 = u_2, y_4 = y_1 + y_2, q_4 = [q_1; q_2]. A_3, B_3, C_3, D_3 are the state-space matrices of G_3 and A_4, B_4, C_4, D_4 those of G_4.

23.a - MC question 23 - 1 points:

Question: Determine the correct matrices A_3 and B_3.

Options: A) A_3 = [[A_1, 0], [0, A_2]], B_3 = [[B_1], [B_2]] | B) A_3 = [[A_1, 0], [B_2 C_1, A_2]], B_3 = [[B_1], [B_2 D_1]] | C) A_3 = [[A_1, 0], [C_2, A_2]], B_3 = [[B_1], [D_2]] | D) A_3 = [[A_1, 0], [B_1 C_2, A_2]], B_3 = [[B_1], [B_1 D_2]]

Key: option B - A_3 = [[A_1, 0], [B_2 C_1, A_2]], B_3 = [[B_1], [B_2 D_1]]

24.a - MC question 24 - 1 points:

Question: Determine the correct matrices A_4 and C_4.

Options: A) A_4 = [[A_1, 0], [B_1 C_2, A_2]], C_4 = [D_1 C_2, C_2] | B) A_4 = [[A_1, 0], [C_2, A_2]], C_4 = [D_1, C_2] | C) A_4 = [[A_1, 0], [0, A_2]], C_4 = [C_1, C_2] | D) A_4 = [[A_1, 0], [B_2 C_1, A_2]], C_4 = [D_2 C_1, C_2]

Key: option C - A_4 = [[A_1, 0], [0, A_2]], C_4 = [C_1, C_2]

25.a - MC question 25 - 1 points:

Question: Select the correct statement.

Options: A) If G_1 has an FIR and G_2 has an IIR then this implies that G_4 has an FIR | B) If G_1 and G_2 have an FIR this implies that G_3 has an IIR | C) Stability of G_1 and G_2 implies stability of G_3 | D) Stability of G_1 and G_2 implies stability of G_4

Key: option D - Stability of G_1 and G_2 implies stability of G_4

26.a - MC question 26 - 1 points:

Question: Assume that you are given the step responses of systems G_3 and G_4: r_3[n] = {..., 0, 2 (n=0), -1, ...} and r_4[n] = {..., 1, 4 (n=0), 3, ...}. Furthermore, the impulse responses of G_1 and G_2 have the forms h_1[n] = {..., 0, 1 (n=0), a, ...} and h_2[n] = {..., 0, 2 (n=0), b, ...}. Determine a and b.

Options: A) a = 1 and b = 2 | B) a = 0 and b = 1 | C) a = -1 and b = 0 | D) a = 0 and b = -1 | E) a = -1 and b = -1 | F) a = -1 and b = 2 | G) a = -2 and b = 1 | H) a = 2 and b = 1

Key: option G - a = -2 and b = 1

Context: Consider the signal x(t) = cos(6 pi t) + sin(8 pi t) + 3.

27.a - MC question 27 - 1 points:

Question: For which of the following sampling times Ts will the signal x(t) not experience aliasing?

Options: A) Ts = 0.5 | B) Ts = 0.1 | C) Ts = 2 | D) Ts = 0.2 | E) Ts = 1 | F) None of the others answers is correct

Key: option B - Ts = 0.1

28.a - MC question 28 - 1 points:

Question: If the signal is sampled with sampling time Ts = 0.2, what is the fundamental period N?

Options: A) N = 15 | B) None of the others answers is correct | C) N = 20 | D) N = 10 | E) N = 1 | F) N = 5

Key: option F - N = 5

Context: Six plots (Plot A to Plot F) show the magnitude of the DFT coefficients |X[k]| of different DT signals, computed using a DFT length of 20 (k = 0..19). [Figures not included.]

29.a - MC question 29 - 1 points:

Question: Let {x_1[n]} = {sin(pi/9 n)}. Which plot shows the magnitude of {X_1[k]}?

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option A - Plot A

30.a - MC question 30 - 1 points:

Question: Let {x_2[n]} = {-cos(3pi/10 n) - 1/2 sin(7pi/10 n) + e^(j 3pi/10 n)}. Which plot shows the magnitude of {X_2[k]}?

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option C - Plot C

31.a - MC question 31 - 1 points:

Question: Let {x_3[n]} = {-1/2 (-1)^n}. Which plot shows the magnitude of {X_3[k]}?

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option F - Plot F

Context: Consider an LTI system described by the transfer function H(z) = 0.5 (z + 1) / (z^2 - 0.5). Define the input signal u[n] = sin(pi/2 n) + cos(pi n) + 2.

32.a - MC question 32 - 1 points:

Question: Determine the DFS coefficients of u[n] (period N = 4).

Options: A) {U[k]} = {8, 2, -2j, 2} | B) {U[k]} = {4, -j, 1, j} | C) {U[k]} = {8, -j, 1, j} | D) {U[k]} = {4, -2j, 2, 2j} | E) {U[k]} = {8, 2j, 2, -2j} | F) {U[k]} = {8, -2j, 4, 2j}

Key: option F - {U[k]} = {8, -2j, 4, 2j}

33.a - MC question 33 - 1 points:

Question: Which of the following plots represents the frequency response H(Omega)? [Answer options are four magnitude/phase plots - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option A - Plot A

34.a - MC question 34 - 1 points:

Question: Determine the DFS coefficients of y[n].

Options: A) {Y[k]} = {16, 2/3 (-j - 1), 8, 2/3 (j - 1)} | B) {Y[k]} = {8, 2/3 (-j - 1), 0, 2/3 (j - 1)} | C) {Y[k]} = {8, 2/3 (-j - 1), 8, 2/3 (j - 1)} | D) {Y[k]} = {16, 2/3 (-j - 1), 16, 2/3 (j - 1)} | E) {Y[k]} = {16, 2/3 (-j - 1), 0, 2/3 (j - 1)} | F) {Y[k]} = {16, 2/3 (j - 1), 0, 2/3 (-j - 1)}

Key: option F - {Y[k]} = {16, 2/3 (j - 1), 0, 2/3 (-j - 1)}

35.a - MC question 35 - 1 points:

Question: If the input signal is u[n] = 1 + cos(pi/2 n), which of the following corresponds to the output signal y[n]?

Options: A) y[n] = 2 + sqrt(2)/3 sin(pi/2 n + 3pi/4) | B) y[n] = 1 + sqrt(2)/4 cos(pi/4 n + 3pi/4) | C) y[n] = 2 + sqrt(2)/3 cos(pi/2 n - 3pi/4) | D) y[n] = 1 + sqrt(2)/4 cos(pi/2 n + 3pi/4) | E) y[n] = 2 + sqrt(2)/4 cos(pi/4 n - 3pi/4) | F) y[n] = 1 + sqrt(2)/3 cos(pi/2 n + 3pi/4)

Key: option C - y[n] = 2 + sqrt(2)/3 cos(pi/2 n - 3pi/4)

Context: Consider six pole-zero plots (Plot A to Plot F) in the complex z-plane, showing the unit circle, poles (x) and zeros (o). [Figures not included.]

36.a - MC question 36 - 1 points:

Question: Which pole-zero plot represents the system with the given frequency response? [Figure: |H(Omega)| - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option A - Plot A

37.a - MC question 37 - 1 points:

Question: Which pole-zero plot represents the system with the given frequency response? [Figure: |H(Omega)| - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option B - Plot B

38.a - MC question 38 - 1 points:

Question: Which pole-zero plot represents the system that produces the given input-output relation? [Figure: |U[k]| and |Y[k]| for k = 0..19 - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option D - Plot D

Context: Consider the signal x[n] = alpha_1 + alpha_2 sin(pi/2 n), with alpha_1, alpha_2 in R+.

39.a - MC question 39 - 1 points:

Question: If alpha_1 = alpha_2 = alpha and the DFS coefficients of x[n] are {X[k]} = {8, -4j, 0, 4j}, what is the value of alpha?

Options: A) alpha = 1 | B) alpha = 2.5 | C) alpha = 3 | D) alpha = 0.5 | E) alpha = 1.5 | F) alpha = 2

Key: option F - alpha = 2

40.a - MC question 40 - 1 points:

Question: If alpha_1 != alpha_2 and the DFS coefficients of x[n] are {X[k]} = {8, -6j, 0, 6j}, what are the values of alpha_1 and alpha_2?

Options: A) alpha_1 = 8, alpha_2 = 6 | B) alpha_1 = 2, alpha_2 = 6 | C) alpha_1 = 8, alpha_2 = 3 | D) alpha_1 = 2, alpha_2 = 4 | E) alpha_1 = 1, alpha_2 = 3 | F) alpha_1 = 2, alpha_2 = 3

Key: option F - alpha_1 = 2, alpha_2 = 3

Context: Consider causal LTI systems G_1 and G_2. G_3 is the cascade u -> G_2 -> x -> G_1 -> y. Separately, the input u[n] is applied directly to G_1, producing output w[n]. The intermediate signal satisfies x[n] = u[n] - u[n-2], and w[n] = s[n] (the unit step).

41.a - MC question 41 - 1 points:

Question: Determine the output signal Y(Omega). [Answer options are four magnitude/phase plots - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option A - Plot A

42.a - MC question 42 - 1 points:

Question: Assume that U(z) = 4 (1 - 0.5 z^(-1)). Determine the frequency response H_3(Omega) of system G_3. [Answer options are four magnitude/phase plots - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option D - Plot D

Context: A continuous-time signal x(t) is sampled with sampling time Ts = 0.001. The DFT of the sampled signal x[n] is computed over 101 samples (k = 0..100). [Figure: |X[k]| - not included.]

43.a - MC question 43 - 1 points:

Question: Which DFT coefficients would you set to zero in order to attenuate the electrical hum at 50 Hz?

Options: A) k = 40 | B) k = 5 and k = 11 | C) k = 50 | D) k = 10 and k = 31 | E) k = 5 and k = 96 | F) k = 10 and k = 91

Key: option E - k = 5 and k = 96

44.a - MC question 44 - 1 points:

Question: Which DFT coefficients would you set to zero in order to filter all the components with frequency higher than 300 Hz?

Options: A) k <= 20 | B) k >= 20 and k <= 81 | C) k >= 30 and k <= 71 | D) k <= 40 | E) k >= 40 | F) k >= 30

Key: option C - k >= 30 and k <= 71

45.a - MC question 45 - 1 points:

Question: Select the correct statement.

Options: A) The magnitude of a causal low-pass Butterworth filter of order 2 rolls off at 60dB/dec. | B) A non-causal filter can have a non-zero phase | C) A weighted low-pass moving-average filter of order 2 has zero phase | D) The magnitude of a causal low-pass moving-average filter of order 2 rolls off at 60dB/dec. | E) A causal filter always has zero phase | F) None of the other statements is correct

Key: option B - A non-causal filter can have a non-zero phase

Context: Consider the following discrete-time LTI filters:
G_A: y[n] = 1/3 u[n] + 1/3 u[n-1] + 1/3 u[n-2]
G_B: y[n] = 1/4 u[n] - 1/4 u[n-1] + 1/4 u[n-2] - 1/4 u[n-3]
G_C: y[n] = 1/2 u[n] - 1/2 u[n-2]
G_D: y[n] = u[n] + u[n-2]

46.a - MC question 46 - 1 points:

Question: Which of the following is a low-pass filter?

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) None of the others

Key: option A - G_A

47.a - MC question 47 - 1 points:

Question: Which of the following is a band-pass filter?

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) None of the others

Key: option C - G_C

48.a - MC question 48 - 1 points:

Question: Which of the following is a high-pass filter?

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) None of the others

Key: option B - G_B

49.a - MC question 49 - 1 points:

Question: Consider a transfer function represented by a pole-zero plot [Figure: pole-zero plot - not included]. What kind of filter does it implement?

Options: A) High-pass filter | B) Low-pass filter | C) None of these | D) Notch filter | E) Band-stop filter | F) Band-pass filter

Key: option D - Notch filter

50.a - MC question 50 - 1 points:

Question: Consider the discrete-time frequency transformation depicted in the figure, where the discrete-time frequency Omega is mapped to the discrete-time frequency Omega_t (both axes from 0 to pi) [Figure not included]. If this transformation is applied to a low-pass filter, what kind of filter is obtained?

Options: A) Band-pass filter | B) None of these | C) Low-pass filter | D) High-pass filter | E) Notch filter | F) Band-stop filter

Key: option C - Low-pass filter

Context: Consider an input signal u[n] with period N = 120 and DFS coefficients U[k] [Figure: |U[k]| for k = 0..119 - not included]. The input u[n] is filtered with a causal moving-average FIR filter of order M with transfer function H_MA(z).

51.a - MC question 51 - 1 points:

Question: Determine the minimum order M that cancels out the input frequency at k = 20.

Options: A) M = 10 | B) M = 9 | C) M = 7 | D) M = 8 | E) M = 5 | F) M = 6

Key: option F - M = 6

52.a - MC question 52 - 1 points:

Question: Assuming that the filter order is M = 3, what is the value of the phase angle H_MA(Omega) at the input frequency corresponding to k = 12?

Options: A) -pi/4 | B) -pi/3 | C) -pi | D) -pi/2 | E) -pi/6 | F) -pi/5

Key: option F - -pi/5

53.a - MC question 53 - 1 points:

Question: What kind of filter is represented by the transfer function H(z) = M/(2M - 1) (H_MA(z) + H_MA(z^(-1)) - 1/M)?

Options: A) Non-causal weighted moving average filter. | B) Causal weighted moving average filter. | C) Non-causal moving average filter. | D) Causal moving average filter.

Key: option C - Non-causal moving average filter.

Context: Consider the first-order continuous-time high-pass filter H(s) = s / (s + w_c) (1), where w_c is the cut-off frequency. The filter is discretized using the bilinear transform, obtaining the first-order discrete-time high-pass filter H(z) = ((1 - alpha) / Ts) (z - 1) / (z + alpha) (2), where alpha = f(Ts, w_c).

54.a - MC question 54 - 1 points:

Question: Determine the function f(Ts, w_c).

Options: A) f(Ts, w_c) = 1 + Ts w_c | B) f(Ts, w_c) = 1 - Ts w_c | C) f(Ts, w_c) = (1 - 0.5 Ts w_c) / (1 + 0.5 Ts w_c) | D) f(Ts, w_c) = (-1 + 0.5 Ts w_c) / (1 + 0.5 Ts w_c) | E) f(Ts, w_c) = (1 + Ts w_c) / (1 - Ts w_c)

Key: option D - f(Ts, w_c) = (-1 + 0.5 Ts w_c) / (1 + 0.5 Ts w_c)

55.a - MC question 55 - 1 points:

Question: Assume that system (1) is sampled with sampling time Ts = 1, and that the desired cut-off frequency for system (2) is Omega_c^des = pi/2. How would you choose alpha?

Options: A) alpha = 0.7 | B) alpha = -0.2 | C) alpha = 0.2 | D) alpha = -0.5 | E) alpha = 0

Key: option E - alpha = 0

56.a - MC question 56 - 1 points:

Question: Which frequency-domain transformation would you apply to system (2) to obtain a band-stop filter?

Options: A) z -> z^2 | B) z -> z^(-1) | C) z -> -z | D) None of the others | E) z -> -z^2

Key: option E - z -> -z^2

57.a - MC question 57 - 1 points:

Question: Consider a band-pass filter with cut-off frequencies w_0 and w_1 [Figure: Bode magnitude |H_BP(omega)| in dB, omega from 10^0 to 10^2 rad/s - not included]. Which of the following low-pass filters could have been used to generate this band-pass filter, obtained after applying the frequency transformation s -> (s^2 + w_0 w_1) / s? [Answer options are four Bode magnitude plots - not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option A - Plot A

58.a - MC question 58 - 1 points:

Question: Consider a periodic input u[n] applied to an LTI filter, producing the output signal y[n] [Figure: u[n] and y[n] for n = 0..100 - not included]. Which is the most likely filter applied to the input?

Options: A) Non-zero phase low-pass filter | B) Non-zero phase high-pass filter | C) All-pass filter | D) None of the others | E) Zero phase high-pass filter | F) Zero-phase low-pass filter

Key: option E - Zero phase high-pass filter

59.a - MC question 59 - 1 points:

Question: Consider the discrete-time system H(z) = z / (z - a), where a in R is an unknown constant. To identify the system, you input u[n] = 1. Unfortunately, you can only measure y[1] = 2. Assuming that there is no noise in the measurement, what is the value of a?

Options: A) a = 2 | B) a = 0.1 | C) It is not possible to identify the system with only one measurement | D) a = 1 | E) a = 0.5 | F) None of the others

Key: option E - a = 0.5

60.a - MC question 60 - 1 points:

Question: Consider the discrete-time system H(z) = z / (z - alpha), identified by measuring its impulse response in the absence of measurement noise. The identified system has the form H_hat(z) = z / (z - alpha_hat). Using N samples gives: N=1: 0, N=2: 0.3333, N=3: 0.4286, N=4: 0.4667, N=5: 0.4839, N=6: 0.4921, N=7: 0.4961, N=8: 0.4980, N=9: 0.4990, N=10: 0.4995. Assuming the true alpha = 0.5, what is the minimum number of samples N_min needed such that |H_Nmin(z)| = |H(z) - H_hat(z)| < 0.5? Hint: H_Nmin(z) is a low-pass filter.

Options: A) N_min = 7 | B) N_min = 2 | C) N_min = 1 | D) N_min = 10 | E) N_min = 5 | F) N_min = 3

Key: option F - N_min = 3